

Eng. Math. - solution of differential equations (DE)
 - equations with derivatives involved

• Falling body: $\frac{d^2 h}{dt^2} = -g$ - 2nd order ordinary (DE)
 • Radioactive decay: $\frac{dA}{dt} = -kA$ - 1st order (ODE) has usual derivative

• Electric circuit: $L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = E(t)$ - ODE - 2nd order

• from psychology: $\frac{y'(t)}{y^{3/2} (1-y)^{3/2}} = \frac{2P}{\sqrt{h}}$ - ODE - 1st order

• wave equation: $\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0$ - 2nd order partial DE (PDE)

Order of DE is the highest order of the derivatives involved

has partial derivatives

Linear DE is DE which contains all derivatives linearly

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1(x) \frac{dy}{dx} + a_0(x) y = F(x)$$

Ex: $\frac{d^2 y}{dx^2} + y^3 = 0$ is not linear

$t^3 \frac{dx}{dt} = t^3 + x$ is linear

$\frac{d^2 y}{dt^2} - \frac{dy}{dt} y = \cos t$ is not linear

Solutions of DE

Def: Function $\varphi(x)$ is explicit solution
of DE $F(x, y, \frac{dy}{dx}, \dots, \frac{d^n y}{dx^n}) = 0$ (1)

if $y = \varphi(x)$ satisfies (1)

Ex: $\frac{d^2 y}{dx^2} - \frac{2}{x^2} y = 0$, $\varphi(x) = x^2 - x^{-1}$

$$\varphi'(x) = 2x + x^{-2} \Rightarrow \varphi''(x) = 2 - 2x^{-3}$$

$$\Rightarrow 2 - 2x^{-3} - \frac{2}{x^2} (x^2 - x^{-1}) = 2 - 2x^{-3} - 2 + 2x^{-3} = 0$$

$\Rightarrow \varphi$ is expl. sol.

$$\varphi(x) = x^3 \Rightarrow \varphi''(x) = 6x \Rightarrow 6x - \frac{2}{x^2} x^3 = 6x - 2x = 4x \neq 0$$

$\Rightarrow \varphi$ is NOT sol.

Def: $D(x, y) = 0$ is implicit sol of (1)
if it gives at least one explicit sol. of (1)

EX: $y^2 - x^2 = 0 \Rightarrow y = \pm x$ If $y = x$ (or $y = -x$)
is expl. sol of (1) then $y^2 - x^2 = 0$ is impl sol
of (1)

EX. $x + y + e^{xy} = 0$ (*) sol of $(1 + xe^{xy}) \frac{dy}{dx} + 1 + ye^{xy} = 0$?

diff. the solution: $\frac{d}{dx}(x + y + e^{xy}) = 0$ (**)

$$\Rightarrow 1 + \frac{dy}{dx} + e^{xy}(xy)' = 0 \Rightarrow 1 + \frac{dy}{dx} + e^{xy}(y + x \frac{dy}{dx}) = 0$$

$$1 + \frac{dy}{dx} + e^{xy}y + xe^{xy} \frac{dy}{dx} = 0$$

$$(1 + xe^{xy}) \frac{dy}{dx} + 1 + ye^{xy} = 0 \leftarrow \Rightarrow (*) \text{ is } \text{impl. sol of } (**)$$